$$\psi_1(x) = 2i\omega x e^{-\frac{\omega}{2\hbar}x^2},$$

or

$$\psi_1(x) = 2i\omega x \psi_0(x).$$

The only important difference between ψ_0 and ψ_1 is the presence of the factor x in ψ_1. This has an effect: it causes the wave function of the first excited state to have a zero, or node, at $x = 0$. This is a pattern that continues as we go up the ladder: each successive excited state has an additional node. We can see this pattern emerge by calculating the second excited state at $n = 2$. All we have to do is apply $\mathbf{a}^+$ again:

$$\psi_2(x) = i\Big(-\hbar\frac{\partial}{\partial x} + \omega x\Big)\Big(xe^{-\frac{\omega}{2\hbar}x^2}\Big).$$

We can see right away that the ωx term will result in an ωx^2 term. The $-\frac{\partial}{\partial x}$, meanwhile, will result in two terms because of the product rule for derivatives. One of these terms will come from the exponential (producing another ωx). The other will come from taking the derivative of x. It's clear that what we'll end up with is a quadratic polynomial. If we work out these derivatives, the resulting wave function is

$$\psi_2(x) = (-\hbar + 2\omega x^2)e^{-\frac{\omega}{2\hbar}x^2}.$$

And so it goes, all the way up the ladder. We can see another pattern here: each eigenfunction is a polynomial in x multiplied by $e^{-\frac{\omega}{2\hbar}x^2}$. Because the exponential goes to zero faster than any of these polynomials grows, each eigenfunction approaches zero asymptotically as x goes to plus or minus infinity. Also, because the degree of each polynomial is one greater than the degree of the previous one, each eigenfunction has one more zero than the previous one.[4] This also explains why successive eigenfunctions alternate